Multimodal Simon Says

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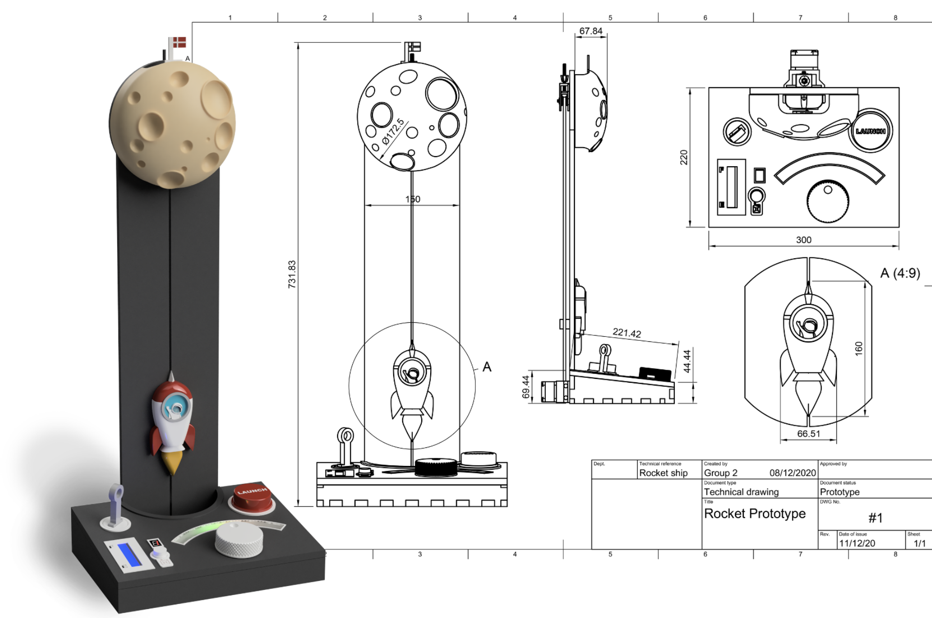

Fig. 1. XXXXXX

ABSTRACT
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1 INTRODUCTION

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2 RELATED WORKS

XXXXXX [1]

3 PROTOTYPE DESIGN

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4 MAIN COMPONENTS

The prototype is split into two power rails. One of these is solely for the vibration motors, which work in a, relatively, narrow voltage range. In addition to this, we also have a separate power source[1] for the motors, to avoid electrical interference, which could negatively impact the microcontrollers.

[1]LiPo battery

The other power rail is directly connected to another LiPo battery, since the compoenents connected to this have a much wider operating voltage range. This means we avoid the efficiency loss from doing a voltage step-up. This means our power rails are, respectively, 3.4 volts[2] and 3.7 volts[3].

4.1 LiPo battery

As mentioned, we use two LiPo batteries each of which has a capacity of 1200 mA-hours. They have an operating voltage range between 3.0 and 4.2 volts, at respectively fully discharged and charged states. Charge time for a completely drained battery is about four hours (TODO: ADD REFERENCE). They are also very reliable and safe, passing all manufacturer safety tests without fire or explosion. This makes them well suited for a toy.

4.2 ATmega328P

The ATmega328P is used in our two agent boards, and control the actuators of the game. They also send information about button presses back to the controller upon request. We make use of eight of their I/O pins directly for the buttons and actuators, and we also make use of their I^2C capabilities to communicate with the controller board. These work in a wide voltage range and have a fairly low power draw.

Due to the simplicity of what we're using them for, and the reduced power draw, we've decided against running these at 16MHz using external crystals. This allows us to avoid stepping up the voltage, since running the chip at 8MHz is stable within the voltage range of a standard LiPo battery as seen in figure 2.

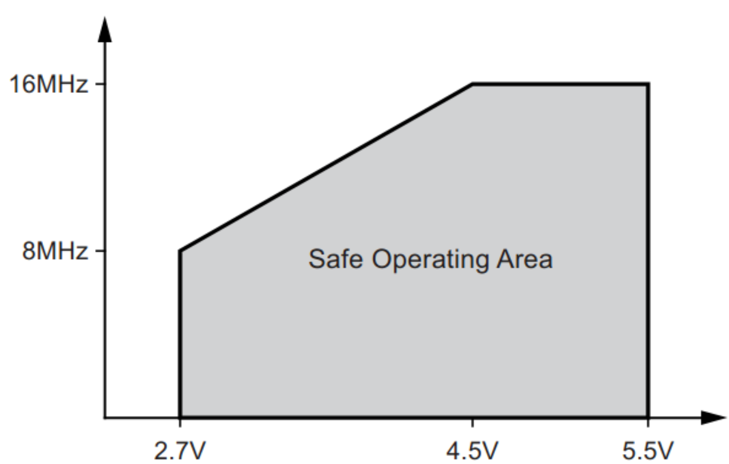

Fig. 2. Safe operating frequencies of the ATmega328P TODO: ADD REFERENCE

To be more specific, we can support voltages between 2.7 and 5.5 volts, which is well within the range of the 3.7 volt LiPo battery. At the voltage we expect the microcontrollers to operate at for the majority of the time, their power draw should be about 3.1mA according to the datasheet (TODO: ADD REFERENCE).

[2] See operating voltage of vibration motors.
[3] Actually a voltage range since it's connected directly to the battery.

4.3 ESP 32

We use the ESP32-C3 super mini in our prototype as the controller of the game. The mictrocontroller requests information from the ATmega328P microcontrollers using I^2C, and sends out appropriate commands. We use the hardware randomness module within the ESP32, to generate sequences for the game. This ensures a reliable and unpredictable pattern. While the operating voltage of the ESP32 itself is only 3.0 to 3.6 (TODO: ADD REFERENCE), the module we're using has a power input that ranges from 3.3 to 6.0 volts (TODO: ADD REFERENCE). Finding an exact expected power draw in the datasheet proved somewhat difficult due to the many possible configurations of the ESP32, but we have concluded it to be between 17 and 28 mA.

4.4 Vibrating Motor

The vibrating motors are used for tactile instructions for the player. They operate at between 2.5 and 3.8 volts, which might lead one to think it could be run directly using the LiPo battery. However, the LiPo batteries actually reach up to 4.2 volts when fully charged, as mentioned, which would go over the limit of 3.8 volts. We therefore decided to step this down to 3.4 volts. The reason for this specific voltage will be explained in section 4.7. It has a maximum rated current draw of 75 mA.

4.5 Bipolar junction transistor

To supply enough current to the vibrating motors, we can't just draw from a pin, since those are only rated for about 40 mA. We want more current than this, so we use the BC547, which has a maximum collector current of 100 mA. This means our vibrating motors fit nicely within the current limit. The voltage rating is also more than sufficient, with both the collector-base and collector-emitter voltage rating being over 40 volts. The emitter-base voltage rating is 6 volts, which is well above our 4.2 maximum voltage.

4.6 Light Emitting Diode

For the LEDs we simply used the ones at the lab, so we don't have the exact datasheet. This isn't strictly necesary though, as LEDs of this size are somewhat standard. We use both the standard single color LEDs and the multicolor LEDs. For the multicolor LEDs, we mostly found that they have a current rating of 20mA (TODO: Add reference) per color. The forward voltage depends on the color, but for our usage, it should be respectively 2V and 3.7V for the red and green colors.

For the single color LED, we've used a standard datasheet we found on a reputable vendor site, and concluded the rated current to be at most 30 mA, but we'll be a bit cautious and say 20mA, as that was the most common we could find for this type of LED. For the forward voltage it, again, depends on color, but we've chosen to assume a forward voltage of 2V, as that will give a greater degree of caution than assuming a higher forward voltage. In practice, we've found the LEDs to be bright enough with these assumptions.

4.7 Low Dropout Regulator

To reduce the voltage from the LiPo to 3.4 volts, we use a low dropout regulator. These are a type of linear voltage regulator, with the advantage that they work even when the input voltage is close to the configured output voltage. This is exactly our use-case. Specifically, we're using the MIC29302WT, which has an output voltage range between 1.25 and 25 V, and an input voltage range between 2.3 to 26 V. The rated current is up to 3A, which is way more than needed for this project, but since the component was bought by a person in the group, they wanted the component to be very versatile for their other projects, which is also the reason for the adjustable output voltage. A more suited component for this prototype could've been something like the (TODO:

INSERT OTHER LDO HERE), which has a lower current rating and outputs 3.3 volts.
The reason for choosing 3.4 volt over 3.3 volt, was to keep the vibration strength higher for longer. While the LDO can maintain the voltage, it does some dropout. According to the datasheet (TODO: ADD REFERENCE), for example, the output voltage might drop as low as 3V, when the battery has reached 3.5 volts, given a high enough current. Though the current won't be that high in our case. To ensure proper vibration, we've therefore set the target voltage a bit higher than we actually need, to hopefully ensure a better player experience for a wider voltage range.

The efficiency of the LDO is simply the output voltage divided by the input voltage, since it's a type of linear voltage regulator. This gives us an efficiency of about $\frac{3.4}{4.2} \approx 0.81$ in the worst case[4].

5 LIMITATIONS AND FUTURE WORK

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6 CONCLUSION

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REFERENCES

[1] Tom Djajadiningrat, Stephan Wensveen, Joep Frens, and Kees Overbeeke. 2004. Tangible products: redressing the balance between appearance and action. *Personal and Ubiquitous Computing* 8 (2004), 294–309.

[4]The worst case efficiency is a fully charged battery.